

Estimating spatial firm productivity spillovers using UK microdata

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DECLARATION

I, Aloysius Lip, hereby declare that the work presented in this dissertation is my own original work. Where information has been derived from other sources, I confirm that this has been clearly and fully identified and acknowledged. No part of this dissertation contains material previously submitted to the examiners of this or any other university, or any material previously submitted for any other assessment.

I declare the use of GenAI tools in an assistative role (UCL Category 2) for this submission. I used Gemini 3.10 Pro (Google DeepMind, 2026). This is limited to the roles of: supporting the particular process of reviewing and refactoring code (data analysis), as a support tutor (model and empirical approach).

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Date: 2nd September, 2026

CLASSIFICATION

This piece of research is primarily (please indicate as appropriate):

- ☒ an empirical/econometric study
- ☐ the examination of a theoretical problem
- ☐ a critical analysis of a policy issue
- ☐ an analytical survey of empirical and/or theoretical literature

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ABSTRACT

~300 word summary of your research topic, methodology and findings

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(if applicable)

Word count:

1. INTRODUCTION

2. LITERATURE

My investigation is rooted in Marshall (1920)'s taxonomy, which identifies three channels to productivity spillovers: (1) input-output costs, (2) labour pooling, and (3) knowledge spillovers. There is no reason to think these are exhaustive. We might theorise that there may be competitive effects and cultural factors. Krugman also labels all these channels as 'centripetal' to an urban core; there are likewise 'centrifugal' factors that push firms away from the urban core, such as destructive (zero-sum) competition, congestion, and saturation costs. A further extension could develop models on these further channels and empirically test them. Furthermore, these productivity spillovers are incredibly salient to thinking about production structures.

In this investigation, I concentrate on (3) knowledge spillovers, tackling a difficult identification problem highlighted in Glaeser (2001). This is more relevant to the knowledge-based services economy that dominates the UK. Input-output costs and labour-pooling matter more for industrial manufacturing firms, especially the latter which was devised in response to the phenomenon emergence of industry towns.

Most agglomeration literature bundles all three channels together. Glaeser (2010) and Krugman (2010) identify how vital identifying knowledge spillovers are. That need to estimate them well only grows with the modern, knowledge economy of the UK. (Goldin, 2023) speculates that agglomeration effects have "increasing strength" attributable to the way in which "knowledge workers in particular become more productive when located near to one another. Information may be more available than ever, but the evidence suggests that it still tends to flow largely through in-person networks".

Specific approaches to investigating knowledge spillovers via innovation, instrumenting via patent publications comes from US evidence, including (Jaffe, 1993), (Berkes and Ruben, 2021) and more recently (Bloom, Schankerman and Van Reenen, 2013).

Knowledge spillovers have been a well-evidence factor of within-firm managerial-derived productivity gains. A field-starting MIT study from Allen (1977) observed an exponentially declining curve depicting the frequency of communication between two individuals against their physical distance in an office.

3. DATA

3.1. FAME

I have used data from Fame for this investigation. Fame is a comprehensive set of financial datapoints from UK firms, with instruments appropriate for this analysis, in particular, number of employees and balance sheet data from which analysis-relevant aggregates can be derived. Fame is provided by Moody's Analytics, an international credit agency, and collects its data from UK firms via survey⁴.

Its international counterpart, Orbis, offers a similar service for international counterparts. This analysis could therefore be extended to offer insights comparing the spillover effects of UK firms to their international counterparts. For this investigation, the UK dataset is comprehensive and the most policy-relevant for this investigation.

Raw data was accessed from Fame in February 2025 by Professor Lars Nesheim for (Irastorza-Fadrique et al., 2026). and reused for this project. Raw data is provided on a series of a financial balance sheet indicators, including characteristics that are fixed for each firm, bucketed into *a1_ID* and *a5_misc* indicator tables, and annual data published on a calendar-year basis, bucketed into *a2_key_finance*, *a3_assets*, and *a4_profits* tables. I designed a series of Python scripts to ingest this data and store it in a 'DuckDB' database, separated into fixed and yearly datapoint tables. 'DuckDB' is an open-source, local, relational (SQL) architecture, designed for processing bulk queries across columns of variables, making it well-suited for econometric data analysis.

3.2. INGESTION AND CLEANING

(...)

All data is deflated to real terms, using 2024 as the base year. I used the GDP deflator for all balance sheet variables, except for *renumeration_employees* and its composites for which I used CPIH, as these are income-related aggregates with real spending power best captured by household purchasing baskets⁵.

All balance sheet data are in thousands ('000s) of GBP (£). As we estimate log relationships exclusively, this has no effect on parameter interpretation.

3.3. LOCATION DATA

3.4. DERIVING PANEL AGGREGATES

Labour input is taken from the number of employees reported on the FAME balance sheet per firm. This dataset, like all others, does not include hours worked per employee on a firm-level that would offer a more granular productivity (GVA per hour worked) measures. An estimation in this setting would likely have to impute the measure from industry aggregates, which would likely confound the true firm-level signal.

Capital is Fame's *fixed_total* variable, which represents the sum of tangibles, intangibles, and other investments within the balance sheet, namely an extremely comprehensive measure of the firm's fixed capital. The average capital-output ratio is 6x within FAME when filtering for

⁴Moody's Analytics (2025)

⁵Office for National Statistics (2026)

positive values Table (I), which is moderately higher than macroeconomic aggregates typically suggest likely due to accounting standard differences.

TABLE I
Capital-output ratios in FAME corpus of firms

Variable	Description	Capital-output ratios
tangibles		3.53
fixed_other	tangibles + intangibles + inv._other	6.00
current_assets	fixed_total + current_assets	11.56

GVA is calculated as the sum of *ebitda* and *renumeration_employees*. *renumeration_employees* is the exact sum of *wages*, *social_security_costs*, and *pensions_costs* in the balance sheet. *ebitda* is in turn *turnover* minus intermediate inputs and is given in the Fame balance sheet for the vast majority of panel observations in our dataset. It is also possible to reconstruct *ebitda* via the sum of *profit_loss_pretax*, *interest_paid*, and *depreciation*, providing an alternative measure of GVA. We can expect the difference between these two to be accounting differences⁶ and additionally note that the alternate GVA measure also reduces panel observations from 1,085,520 to 618,588. For these reasons, I do not report TFP calculations with this second GVA measure which are regardless near-identical, though I resultingly refer to GVA throughout this paper as *gva_1*.

3.5. FIRM DISTANCE MEASURES

Throughout this analysis, I use terminology of a productivity spillover occurring from a group or average of *peer* firms onto a *target* firm. This relationship is irreflexive. For groups it is transitive: if firms A and B share a spatial group, such as a travel-to-work area (*ttwa*), first-half of a UK postcode (*pc4*), or full postcode (*pc8*), then a firm C that is a peer of firm B must also be of firm A. For continuous distance measures, this is non-transitive. Firm B, a peer of firm A within its 5km radius, can clearly have its own peers that are within its own 5km radius that are not shared with A. Every firm is then its own overlapping group, or equivalently, a sparse, rather than block diagonal, firm-to-firm matrix⁷.

(Cleaning address data).

Fame only reports each company’s last-known registered and trading address, at the time of compilation or publication of its annual vintage(cite). As a result, final location data is only available at the firm level. We can interpret each firm as being static over the entire 20-year sample, with no movers. This seems very unlikely (expand). Instead, we must interpret the constructing address, and therefore firm-to-firm distance measures as a signal of firm pairs’ true distance polluted by movers. This has 2 econometric consequences.

Firstly, firm fixed effects in all spatial specifications become extremely important. For all movers within the sample period, their firm location decision is endogenous to the subsequent productivity spillovers they experience and exert, for reasons discussed in Section 2 on New Economic Geography. (More on firm FEs) There is no solution to this for moves occurring out-of-sample, for instance, just before 2009.

Secondly, I acknowledge some attenuation to the productivity spillover estimate Peer firms

⁶An OLS regression between the two, discarding 5% outliers, has a coefficient of 0.91 with a 0.000 p-value. This means *gva_2* is slightly lower than *gva_1*, with that impact mostly driven by top-end outliers.

⁷I do actually impose a block diagonal structure for reasons of computation as explained subsequently

who were close to the target firm during the firm-year observation who subsequently moved away within-sample will have an extreme distance recorded, and so be excluded from tagged address groups and truncated distance calculations. Peer firms who are recorded as close to the target firm as they subsequently moved to the location (More on direction of bias/positive selection?)

I first consider a model where each target has peers which share a spatial identifier, meaning they belong to the same travel-to-work-area (*ttwa*) using the 2011 definitions ⁸, the first half of a UK postcode *pc4*, and the full UK postcode *pc8*.

(Discuss and illustrate different forms of ω , with illustrative block diagonal matrices)

⁸(Review ONS bulletin on TTWAs data source and describe clearly)

4. MODEL

I use a stylised 3-factor production model throughout this analysis. Y , L , K are each firm-year observation's total GVA, labour, and capital as usual. Z_{jt} is the 2-factor total-factor productivity (TFP) residual. I use lower-case variables to indicate logged terms and bolded variables to indicate stacked vectors.

$$Y_{it} = A \cdot L_{it}^\alpha K_{it}^\beta \left[\prod_{j \neq i} f(Z_{jt}) \right]^\gamma \quad (1)$$

$$y_{it} = c + \alpha l_{it} + \beta k_{it} + \gamma \sum_{j \neq i} \ln f(Z_{jt}) \quad (2)$$

$$z_{it} = y_{it} - c - \alpha l_{it} - \beta k_{it} \quad (3)$$

$$f(Z_{jt}) = Z_{jt}^{w_{ijt}} \quad (4)$$

$$z_{it} = c + \gamma \sum_{j \neq i} w_{ijt} z_{jt} \quad (5)$$

$$\mathbf{z}_t = c + \gamma \mathbf{\Omega}_t \mathbf{z}_t \quad (6)$$

Theory suggests no restrictions on α , β , and γ collectively should be imposed, as γ is a positive externality. (Lucas discusses this)

In equation 4 I assume log-linearity for the peer TFP effect on the target, namely the $f(\cdot)$ transformation function dictating effects for peers contemporaneously is a Cobb-Douglas weighted set of indices, s.t. $\sum_{j \neq i} w_{ijt} = 1$. This permits me to estimate structural γ in two stages should I require, namely using y_{it} log GVA as the regressand is equivalent to first estimating z_{it} via a 2-factor TFP estimation and then secondly regressing z_{it} against its peers.

To estimate structural γ , I use 3 linear models, increasing in complexity, each resolving some of the endogeneity and identification issues of the last. Though the final model I present has the most accurate interpretation of parameters, earlier models are useful for iteratively understanding the data and making choices about parameter selection along the way since they are computationally easier to calculate and assuming that relationships between these parameters are similar. Choosing all these parameters once, in a final model, would have required a large amount of computation via cross-validation, which this investigation's limits on time and computation did not permit.

TABLE II
(Summary of econometric models)

Model	Stages	Identification	Endog.	Consistent?	Decisions
1. LMM grouped peers	2				
2. Static distance decay	2				
3. Dynamic panel	1				

'Stages' indicates whether the derived 2-factor TFP value z_{it} was used as the regressand, or the model used y_{it} from the *gval* variable directly.

4.1. 2-FACTOR TFP ESTIMATION

As established above, it is helpful to conduct a first stage derividing the Solow residual, z_{ijt} , from a 2-factor production model, which I will analyse and decompose further in a second stage. This measurement is only helpful for static models, so the spillover from peer firms via Z_{jt} only occurs in the same t -period. As I will show, there may be a dynamic lag from peer TFP z_{jt} to target TFP z_{it} , so for the later model, z_{it} needs to be estimated in full in a single-stage with dynamic variables.

$$\begin{aligned} y_{it} &= \alpha_i + \gamma_t + \beta_0 + \beta_1 l_{it} + \beta_2 k_{it} + \varepsilon_{it} \\ z_{it} &= y_{it} - c - \alpha l_{it} - \beta k_{it} \end{aligned} \tag{7}$$

I conduct the reduced-form static panel regression in logs, with firm and time fixed effects. I assume exogeneity between the residual and factor input levels.

TABLE III
Baseline TFP calculation from 2-factor production

	(1)	(2)	(3)	(4)	(5)
	<i>gva</i> ₁	<i>gva</i> ₁	<i>gva</i> ₁	<i>gva</i> ₁	<i>gva</i> ₁
constant	3.201 (0.032)***	4.840 (0.019)***	3.223 (0.027)***	1.994 (0.010)***	1.996 (0.010)***
$\ln(\textit{employees})$	0.621 (0.005)***	0.717 (0.005)***	0.618 (0.004)***	0.559 (0.002)***	0.559 (0.002)***
$\ln(\textit{fixed_total})$		0.058 (0.001)***			
$\ln(\textit{total_assets})$	0.265 (0.004)***		0.264 (0.004)***	0.422 (0.002)***	0.422 (0.002)***
<i>i</i> fixed effects	✓	✓	✓		
<i>t</i> fixed effects	✓	✓		✓	
Observations	1,081,520	1,042,854	1,081,520	1,081,520	1,081,520
R ²	0.3701	0.3226	0.3861	0.8345	0.8346

All parameters reported with standard errors underneath. Significant estimates against an $H_0 : \beta = 0$ are indicated with $p < 0.1$ (*), $p < 0.05$ (**), and $p < 0.01$ (***).

The parameter estimates give a labour income share exactly in line with ONS analysis of UK industry-level data over the same time period, particularly *corporate gross labour share* which is constructed from a similar sample to the coverage in Fame⁹. Specifications without fixed or time effects offer similar parameter estimates. The capital share of income jumps on a specification with only time fixed effects, which is to be expected (explain).

Oddly, running the regression with *fixed_total* as the capital measure instead of *total_assets* almost entirely eliminates the capital share of income from the regression. For the rest of the analysis, I use *total_assets* as the primary capital measure and leave it for future analysis why this dataset yields this phenomenon.

TFP is the residual on GVA after extracting the proportion that can be accounted by capital and labour inputs. Equivalently, it is the sum of fixed effects in this specification with the error term, which means we should expect it to be exogenous to capital and labour inputs,

⁹Office for National Statistics (2024)

but correlated to both firm effects and time trends. By extracting this regression residual, I have a TFP measure for all of my 1,081,520 panel observations.

4.2. LMM GROUPED PEERS

Section 3.5 discusses how firms have been labelled to symmetric, transitive, non-overlapping groups by *pc8* (smallest), *pc4*, and *ttwa* (largest). I conduct a linear regression using the averages of peers belonging to these groups, leaving out the target firm. This is known as the 'Linear-in-Means (LMM)' Model and is common to the peer effects literature.

$$z_{it} = \alpha_i + \gamma_t + \beta_0 + \sum_{g=1}^3 \beta_g \mathbb{E}[z_{jt}|g] + \varepsilon_{it} \quad (8)$$

$$\begin{aligned} \mathbf{z}_t &= c + \gamma \mathbf{\Omega}_t \mathbf{z}_t \\ w_{ijt} \in \mathbf{\Omega}_t &= \begin{cases} 1, & \text{if } (i, j) \in g \\ 0, & \text{if } i = j \\ 0, & \text{otherwise} \end{cases} \end{aligned} \quad (9)$$

It is crucial that I have multiple, non-overlapping groups in a single model, where my target firm z_{it} is a member of all groups simultaneously, but peers in one group are excluded from all others.

The non-overlapping property prevents multicollinearity. If I were to use nested, overlapping geographies, namely keeping all firms within the *ttwa* area in that group average, including the ones I have just counted in the *pc4* and *itxpc8* groups, then my group averages mechanically align: $z_{jt,pc8}$ is embedded within $z_{jt,pc4}$, and both are within $z_{jt,ttwa}$, making β_g parameter estimates unstable.

The single model approach prevents omitted variable bias (OVB). $\mathbb{E}[z_{jt,ttwa}|g] \propto \mathbb{E}[z_{jt,pc8}|g]$, so running three models where the former is just my single regressor means the latent $\beta_{pc8} \in u_{it}$ is correlated with both z_{itg} and z_{jtg} . We could reasonably argue that the inner rings are causal, so this causal factor excluded from the regression influences both and creates heavily upward bias on my outer-ring estimations. Indeed, this critique extends within further inner rings. If I think the causal unit is not peer firms within the *pc8* group at all, but a further hyper-local effect - perhaps those on the same street or same building, then this latent factor will correlate with both my outcome and regressor and drive the results. However, given the small number of peers contained within an *pc8* unit and the limits of data availability to identify firms at more local granularities, I proceed assuming the effect from *pc8* peers is homogenous.

Finally, the approach using multiple groups as regressors at once gives identification by breaking transitivity. If I were to run the same regression with merely one group, say *pc8*, then my regression would suffer the reflection problem outlined in (Manski, 1993). As I am estimating the expected value of z_{it} and z_{jt} from the same sample, they cannot be uniquely identified in reduced form.

Reflection problem with a single regressor, ι denotes a column vector of 1s

(review: maths is a bit tricky here)

$$\begin{aligned}
\mathbb{E}[z_{itg}|g] &= \alpha_i + \gamma_t + \beta_0 + \beta_1 \mathbb{E}[z_{jtg}|g] \\
&= (\alpha_i + \gamma_t + \beta_0)\iota + \beta_1 \mathbf{\Omega}_t \mathbb{E}[\mathbf{z}_{tg}|g] \\
&= (\mathbb{I} - \beta_1 \mathbf{\Omega}_t)^{-1}(\alpha_i + \gamma_t + \beta_0)\iota
\end{aligned}$$

This problem is mitigated with having 3 distinct groups to which z_{it} belongs. The results of the LMM estimation implementing equation 8 are below:

TABLE IV
Static panel linear-in-means using spatial-group TFP averages

	(1) <i>tfp</i>	(2) <i>tfp</i>	(3) <i>tfp</i>
constant	2.075 (0.044)***	0.310 (0.038)***	1.202 (0.215)***
$\ln(\text{employees})$			0.184 (0.044)***
<i>peer_tfp_pc8</i>	0.280 (0.005)***	0.559 (0.004)***	0.280 (0.005)***
<i>peer_tfp_pc4_donut</i>	0.015 (0.005)***	0.142 (0.008)***	0.015 (0.005)***
<i>peer_tfp_ttwa_donut</i>	0.033 (0.013)***	0.204 (0.013)***	0.306 (0.071)***
<i>peer_tfp_pc8_x_employees</i>			-0.058 (0.015)***
<i>i</i> fixed effects	✓		✓
<i>t</i> fixed effects	✓	✓	✓
Observations	764,537	764,537	764,537
R ²	0.0388	0.1792	0.0390

The parameter results suggest a strong correlation between the TFP of a target firm and its peers.

4.3. STATIC DISTANCE DECAY

To improve on the LMM specification, I make a modification where every peer firm is weighted by its influence on the target by the reciprocal of the continuous distance metric calculated in 3.5 $w_{ijt} = f(d)$. This is stylistic of a declining peer influence further away, representing a (lower frequency of social collisions, barriers created via transport and time costs, and an increased difficulty of observation of management practices).

$$z_{it} = \alpha_i + \gamma_t + \beta_0 + \beta_1 \sum_{j \in -i} w_{ijt} z_{jt} + \varepsilon_{it} \quad (10)$$

$$\mathbf{z}_t = (\alpha_i + \gamma_t + \beta_0)\mathbf{1} + \beta_1 \sum_{j \in -i} \Omega_t \mathbf{z}_t + \varepsilon_t \quad (11)$$

$$w_{ijt} \in \Omega_t = \begin{cases} \frac{1}{d_{ij}}, & (1) \text{ gravity} \\ \frac{1}{d_{ij}^2} & (2) \text{ N.E.G.} \\ \exp -d_{ij} \cdot k & (3) \text{ exponential decline} \end{cases} \quad (12)$$

This represents a likely more accurate declining effect of productivity spillover, rather than the special-case group mean where every firm in an enclosed area, for example, the target's postcode, has an equal influence on the target's TFP. In addition, it allows for better identification following Bramoullé et al. (2009). (explain Bramouille2009, patterns of) Ω_t

In essence, the LMM specification only contains 3 non-overlapping groups from its location tags. The distance-decayed specification contains as many non-overlapping groups as firms.

TABLE V
Static panel using continuous distance-decay weighted
TFP average of peers

	(1) <i>tfp</i>	(2) <i>tfp</i>	(3) <i>tfp</i>
constant	2.479 (0.029)***	2.586 (0.029)***	2.417 (0.032)***
$\ln(\textit{employees})$	-0.009 (0.004)**	-0.009 (0.004)**	-0.009 (0.004)**
$\ln(\textit{nb_peers})$	0.013 (0.004)***	0.012 (0.004)***	0.016 (0.004)***
<i>tfp_wav</i> ₁	0.168 (0.003)***		
<i>tfp_wav</i> ₂		0.135 (0.003)***	
<i>tfp_wav</i> ₃			0.184 (0.005)***
<i>i</i> fixed effects	✓	✓	✓
<i>t</i> fixed effects	✓	✓	✓
Observations	1,067,013	1,067,013	1,067,013
R ²	0.0101	0.0095	0.0040

This distance-decay specification revises down the estimate from peers to a an elasticity of 0.168, lower than the results given by the LMM specification. This is likely due to (...)

In addition, I introduce controls (...)

4.4. DYNAMIC PANEL

This analysis rests on the plausibility of the group peer effect, $\sum_{j \in -i} w_{ijt} z_{jt}$ as an exogenous regressor. To support this argument, I introduce a dynamic panel specification, such that lagged realisations of the regressand are used as additional regressors. I can implement these lags exploiting the rich time series observations available in Fame.

Autoregressive lags help control for reverse causality: if $y_{i,t-k}$ is causal for z_{jt} , or feasibly $z_{j,t-1}$ and $\mathbb{E}[z_{j,t-1}, z_{j,t}] \neq 0$, then the presence of $y_{i,t-k}$ as a control helps eliminate that bias. Similarly, theory offers us no clear view on where there could be simultaneity, where variation in $\Delta y_{i,t}$ is driven causally by $\Delta y_{i,t-1}$ and that change in $\Delta y_{i,t}$ in turn drives $\Delta z_{j,t}$. The autoregressive regressor helps control for that. Additionally, the results below suggest the autoregressive parameter coefficient is low, with a log elasticity of 0.005 (significant at the 1% level) in all specifications, suggesting any concerns about reverse causality or simultaneity driven by an AR process are likely minimal. The use of this dynamic regressor improves the plausible exogeneity of $z_{i,j,t}$.

The autoregressive specification requires a return to the one-stage production function with an outcome of GVA, y_{it} . I continue to use the scalar weighted distance TFP metric as the primary regressor of interest z_{jt} .

$$y_{i,t} = \alpha_i + \beta_0 + \sum_{k=1}^p \gamma_k y_{i,t-k} + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \beta_3 \sum_{j \in -i} w_{i,j,t} z_{j,t} + \varepsilon_{i,t} \quad (13)$$

$$\Delta y_{i,t} = \sum_{k=1}^p \gamma_k \Delta y_{i,t-k} + \beta_1 \Delta k_{i,t} + \beta_2 \Delta l_{i,t} + \beta_3 \Delta \sum_{j \in -i} w_{i,j,t} z_{j,t} + \Delta \varepsilon_{i,t} \quad (14)$$

I list an AR(p) specification above, however, my results show that AR(1) is the true structure. Appendix I, which increases the order using a nested model approach, shows that for any specification order above an AR(1) coefficients become insignificant suggesting multicollinearity (F-test on AR(2)). I also confirm this via information criteria tests, (which show) that lags beyond $t-2$ provide no new information for the variation on $\ln(y_{i,t})$. Noting that the regression coefficient on the $\ln(gva_1)_{t-1}$ is significant at the 1% level in my results, I can strongly infer that an AR(1) is the true structure.

((IC test results))

This dynamic panel approach requires a valid choice of instruments. Since I take first differences of my log aggregates to strip out firm fixed effects, I strongly expect $\Delta \ln(y_{it})$, my regressand, to be correlated with $\Delta \ln(y_{it})$, my AR regressor, meaning this first-order lag should be treated as a pre-determined regressor. A naïve OLS would fail strict exogeneity in this setting, as error exogeneity is more limited.

$$\begin{aligned} \mathbb{E}[\varepsilon_{it} | x_{i\tau}] &= 0 \quad \forall t \geq \tau \\ \delta y_{i,\tau-k}, \delta k_{i,\tau} &\in x_{i,t}, \quad k \leq p \end{aligned} \quad (15)$$

This is confirmed by the Anderson-Rubin (AR(1)) test in my results, which rejects the null hypothesis at the 1% level for all models, suggesting that first-order lags of log GVA are a

weak instrument (Why? Shouldn't they fail exclusion restriction, not instrument correlation). In addition, I assume that capital, $k_{i,t}$ is pre-determined, as investment is chosen one period ahead, in line with standard Perpetual Inventory Method approaches¹⁰ (Review predetermined material. Additionally, explain the logic of instrumenting for capital)

To overcome this endogeneity, I introduce deeper autoregressive lags as instrumental variables, as proposed by (Anderson and Hsiao, 1981). I use 3 autoregressive instrument lags starting from $t - 2$: $\ln(y_{i,t-2}), \ln(y_{i,t-3}), \ln(y_{i,t-p-3})$ for log GVA, and 2 lags for capital $\ln(k_{i,t-p-1}), \ln(k_{i,t-p-2})$. An Anderson-Rubin (AR(2)) test shows that deep lags are a valid instrument for log GVA. The test applied to the second lag of $\ln(gva_1)$ fails to reject the null hypothesis of for any specification, so I assume it is a valid instrument.

Additionally, using multiple deep lags allows for overidentification: in this basic AR(1) specification, I have $k = 4$ regressors and $l = 7$ instruments. I could increase this. Fame is rich on the time dimension, with up to 19 lags at most for a firm, and a median of 6 lags, so I could safely provide more lags of regressors as instruments. Further lags is likely to have a decreasing marginal effect on explanatory power and additionally ((...))

(Discuss introducing a new weighting method using past firm similarity.)

I estimate the equation fulfilling the vector of moment conditions, (16) via GMM.

$$\mathbb{E} \left[\Delta \varepsilon_{it} \cdot \begin{pmatrix} \Delta \ln(y_{i,t-2}) \\ \Delta \ln(y_{i,t-3}) \\ \Delta \ln(y_{i,t-4}) \\ \Delta \ln(k_{i,t-2}) \\ \Delta \ln(k_{i,t-3}) \\ \Delta \Sigma_j w_{ijt} z_{jt} \end{pmatrix} \right] = 0 \quad (16)$$

As my regressors are over-identified, I use 2-step GMM to improve efficiency. (I should introduce a control (basically a diff-in-diff?) of peers from the same industry across the country (outside the spatial bubble).)

Finally, a Sargan-Hansen J-test is implemented to check if these instruments are over-identified, meaning the moment conditions cannot be fulfilled with 95% confidence. In all models, the J-test fails to reject the null hypothesis, meaning the model's moment conditions are jointly correct.

5. DYNAMIC PANEL WITH ADDITIONAL CONTROLS

¹⁰Noting that this is ordinarily done on a quarterly basis, however, due to data limitations we adopt the same thing here even though Fame provides observations annually.

TABLE VI
Dynamic panel using instrumented continuous
distance-decay weighted TFP

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$
constant	1.371 (0.134)***	1.577 (0.144)***	1.064 (0.103)***
$\ln(gva_1)_{t-1}$	0.005 (0.001)***	0.005 (0.001)***	0.005 (0.001)***
$\ln(employees)$	0.696 (0.038)***	0.689 (0.037)***	0.709 (0.040)***
$\ln(total_assets)$	0.259 (0.041)***	0.266 (0.041)***	0.247 (0.044)***
tfp_wav_1	0.503 (0.028)***		
tfp_wav_2		0.423 (0.023)***	
tfp_wav_3			0.618 (0.042)***
Hansen Test	X (0.417)	X (0.459)	X (0.246)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.746)	X (0.600)	X (0.945)
i fixed effects	✓	✓	✓
t fixed effects	✓	✓	✓
Observations	713,339	713,339	713,339

Model includes a constant as the baseline for t fixed effects.

BIBLIOGRAPHY

REFERENCES

- T W Anderson and Cheng Hsiao. Estimation of Dynamic Models with Error Components. *Journal of the American Statistical Association*, 76(375):598–606, 1981. ISSN 01621459, 1537274X. doi: 10.2307/2287517. URL <http://www.jstor.org/stable/2287517>.
- Yann Bramoullé, Habiba Djebbari, and Bernard Fortin. Identification of peer effects through social networks. *Journal of Econometrics*, 150(1):41–55, 2009. doi: 10.1016/j.jeconom.2008.12.021.
- Google DeepMind. Gemini 3.1 Pro, 2026. URL <https://storage.googleapis.com/deepmind-media/Model-Cards/Gemini-3-1-Pro-Model-Card.pdf>.
- Aitor Irastorza-Fadrique, Lars Nesheim, and Peter Levell. Three Not-so-easy Puzzles : Post Financial Crisis Performance of the UK Corporate Sector. (*forthcoming*), 2026.
- Charles F Manski. Identification of Endogenous Social Effects: The Reflection Problem. *The Review of Economic Studies*, 60(3):531–542, 1993. ISSN 00346527, 1467937X. doi: 10.2307/2298123. URL <http://www.jstor.org/stable/2298123>.
- Moody’s Analytics. Fame, feb 2025.
- Office for National Statistics. Trends in the UK labour share, nov 2024. URL <https://www.ons.gov.uk/economy/economicoutputandproductivity/output/articles/trendsintheuklabourshare1997to2023/2024-11-25#an-overview-of-the-labour-share>.
- Office for National Statistics. Consumer prices indices technical guidance, may 2026. URL <https://www.ons.gov.uk/economy/inflationandpriceindices/methodologies/consumerpricesindices/technicalguidance>.

APPENDIX 1

TABLE VII
Dynamic panel: higher-order autoregressive lags

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$	(4) $\ln(gva_1)$
constant	1.371 (0.134)***	1.446 (0.843)	0.208 (2.578)	0.459 (2.844)
$\ln(gva_1)_{t-1}$	0.005 (0.001)***	0.019 (0.104)	0.128 (0.295)	-0.020 (0.256)
$\ln(gva_1)_{t-2}$		0.002 (0.003)	0.004 (0.101)	0.094 (0.300)
$\ln(gva_1)_{t-3}$			-0.001 (0.006)	0.036 (0.162)
$\ln(gva_1)_{t-4}$				-0.003 (0.004)
$\ln(employees)$	0.696 (0.038)***	0.762 (0.097)***	0.736 (0.149)***	0.659 (0.323)*
$\ln(total_assets)$	0.259 (0.041)***	0.190 (0.090)*	0.248 (0.121)*	0.291 (0.305)
tfp_wav_1	0.503 (0.028)***	0.548 (0.059)***	0.508 (0.076)***	0.466 (0.184)*
Hansen Test	X (0.417)	X (0.922)	X (0.881)	X (0.826)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	X (0.140)	X (0.212)
AR(2) Test	X (0.746)	X (0.925)	X (0.802)	X (0.668)
i fixed effects	✓	✓	✓	✓
t fixed effects	✓	✓	✓	✓
Observations	713,339	603,058	511,396	433,185

TABLE VIII
Dynamic panel: higher-order regressor lags

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$	(4) $\ln(gva_1)$	(5) $\ln(gva_1)$
constant	1.124 (0.113)***	0.956 (0.101)***	0.743 (0.093)***	0.584 (0.091)***	0.543 (0.094)***
$\ln(gva_1)_{t-1}$	0.002 (0.001)*	0.003 (0.001)***	0.005 (0.001)***	0.005 (0.001)***	0.005 (0.001)***
tfp_wav_{1t-1}	0.066 (0.006)***	0.077 (0.007)***	0.069 (0.007)***	0.062 (0.007)***	0.061 (0.008)***
tfp_wav_{1t-2}		0.052 (0.005)***	0.055 (0.006)***	0.050 (0.007)***	0.051 (0.007)***
tfp_wav_{1t-3}			0.043 (0.005)***	0.048 (0.006)***	0.049 (0.006)***
tfp_wav_{1t-4}				0.030 (0.005)***	0.038 (0.006)***
tfp_wav_{1t-5}					0.028 (0.005)***
$\ln(employees)$	0.686 (0.036)***	0.695 (0.038)***	0.670 (0.040)***	0.644 (0.044)***	0.665 (0.052)***
$\ln(total_assets)$	0.270 (0.040)***	0.260 (0.042)***	0.287 (0.045)***	0.317 (0.049)***	0.294 (0.057)***
tfp_wav_1	0.508 (0.028)***	0.514 (0.029)***	0.492 (0.030)***	0.469 (0.033)***	0.480 (0.038)***
Hansen Test	X (0.492)	X (0.968)	X (0.453)	X (0.084)*	X (0.080)*
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.279)	X (0.612)	X (0.801)	X (0.854)	X (0.729)
i fixed effects	✓	✓	✓	✓	✓
t fixed effects	✓	✓	✓	✓	✓
Observations	713,339	603,058	511,396	433,185	365,908

TABLE IX
Dynamic panel: higher-order regressor lags

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$	(4) $\ln(gva_1)$	(5) $\ln(gva_1)$
constant	0.421 (0.082)***	0.388 (0.081)***	0.270 (0.077)***	0.199 (0.083)*	0.116 (0.093)
$\ln(gva_1)_{t-1}$	0.004 (0.001)***	0.004 (0.001)**	0.003 (0.001)*	0.002 (0.002)	0.002 (0.002)
tfp_wav_{1t-1}	0.055 (0.008)***	0.056 (0.008)***	0.047 (0.008)***	0.042 (0.009)***	0.036 (0.011)**
tfp_wav_{1t-2}	0.045 (0.007)***	0.045 (0.007)***	0.040 (0.007)***	0.037 (0.007)***	0.035 (0.009)***
tfp_wav_{1t-3}	0.046 (0.006)***	0.046 (0.006)***	0.039 (0.006)***	0.033 (0.007)***	0.029 (0.008)***
tfp_wav_{1t-4}	0.034 (0.006)***	0.031 (0.006)***	0.024 (0.006)***	0.019 (0.007)**	0.017 (0.008)*
tfp_wav_{1t-5}	0.029 (0.005)***	0.030 (0.006)***	0.023 (0.006)***	0.019 (0.006)**	0.017 (0.006)**
tfp_wav_{1t-6}	0.018 (0.004)***	0.024 (0.005)***	0.021 (0.005)***	0.019 (0.005)***	0.017 (0.006)**
tfp_wav_{1t-7}		0.019 (0.005)***	0.017 (0.005)***	0.015 (0.006)**	0.011 (0.007)
tfp_wav_{1t-8}			0.013 (0.004)**	0.016 (0.005)**	0.017 (0.006)**
tfp_wav_{1t-9}				0.010 (0.005)*	0.013 (0.005)*
$\ln(employees)$	0.628 (0.049)***	0.636 (0.053)***	0.582 (0.055)***	0.548 (0.061)***	0.523 (0.073)***
tfp_wav_{1t-10}					0.016 (0.006)**
$\ln(total_assets)$	0.334 (0.054)***	0.325 (0.058)***	0.384 (0.061)***	0.421 (0.067)***	0.448 (0.080)***
tfp_wav_1	0.452 (0.036)***	0.457 (0.038)***	0.420 (0.040)***	0.397 (0.044)***	0.380 (0.052)***
Hansen Test	X (0.097)*	X (0.122)	X (0.450)	X (0.637)	X (0.435)
AR(1) Test	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***	✓ (0.000)***
AR(2) Test	X (0.924)	X (0.962)	X (0.568)	X (0.507)	X (0.450)
i fixed effects	✓	✓	✓	✓	✓
t fixed effects	✓	✓	✓	✓	✓
Observations	307,394	256,016	210,798	171,125	136,527

TABLE X
Dynamic panel: higher-order regressor lags

	(1) $\ln(gva_1)$	(2) $\ln(gva_1)$	(3) $\ln(gva_1)$	(4) $\ln(gva_1)$	(5) $\ln(gva_1)$	(6) $\ln(gva_1)$
constant	1.164 (0.573)*	0.498 (1.561)	0.398 (1.667)	1.494 (0.428)***	0.333 (1.469)	1.351 (0.307)***
$\ln(gva_1)_{t-1}$	-0.011 (0.117)	0.081 (0.280)	-0.028 (0.251)	-0.129 (0.081)	0.087 (0.276)	-0.125 (0.063)*
$\ln(gva_1)_{t-2}$	-0.000 (0.003)	0.006 (0.111)	0.076 (0.385)	0.004 (0.002)	-0.001 (0.111)	0.006 (0.002)**
$\ln(gva_1)_{t-3}$		-0.001 (0.005)	0.034 (0.156)		-0.002 (0.004)	
$\ln(gva_1)_{t-4}$			-0.002 (0.004)			
tfp_wav_1	0.551 (0.052)***	0.504 (0.078)***	0.475 (0.201)*	0.525 (0.049)***	0.488 (0.073)***	0.524 (0.050)***
tfp_wav_{1t-1}	0.099 (0.102)	0.005 (0.255)	0.082 (0.217)	0.196 (0.073)**	-0.005 (0.249)	0.189 (0.055)***
tfp_wav_{1t-2}	0.060 (0.011)***	0.036 (0.099)	-0.019 (0.342)	0.072 (0.012)***	0.046 (0.102)	0.070 (0.010)***
tfp_wav_{1t-3}				0.056 (0.009)***	0.039 (0.020)	0.065 (0.008)***
tfp_wav_{1t-4}						0.042 (0.006)***
$\ln(employees)$	0.737 (0.099)***	0.720 (0.146)***	0.643 (0.382)	0.650 (0.075)***	0.705 (0.148)***	0.657 (0.079)***
$\ln(total_assets)$	0.209 (0.088)*	0.253 (0.119)*	0.307 (0.367)	0.272 (0.074)***	0.271 (0.119)*	0.266 (0.079)***
Hansen Test	X (0.971)	X (0.855)	X (0.849)	X (0.999)	X (0.851)	X (0.232)
AR(1) Test	✓ (0.002)***	X (0.124)	X (0.163)	✓ (0.000)***	X (0.111)	✓ (0.000)***
AR(2) Test	X (0.889)	X (0.878)	X (0.775)	X (0.092)*	X (0.837)	✓ (0.030)**
i fixed effects	✓	✓	✓	✓	✓	✓
t fixed effects	✓	✓	✓	✓	✓	✓
Observations	603,058	511,396	433,185	511,396	511,396	433,185

APPENDIX 2

APPENDIX 3